

A Hybrid Deep Learning Approach for Automated Plant Disease Diagnosis in Indian Ridge Gourd: Dataset Creation and Real-World Application

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Abstract. Timely and precise detection of plant diseases is a crucial component in ensuring sustainable agriculture and global food security. The traditional diagnostic methods are based on manual inspection, which are often subjective, labour-intensive, and prone to human error, leading to significant yield loss. To overcome these challenges, the present study introduces a deep learning-based system for automating the diagnosis of leaf diseases in the Indian Ridge gourd, a crop of local importance that remains underexplored. The current study began with the creation of a, real-world dataset containing 600 ridge gourd leaf images collected from farms in Vidhani, Jaipur, India. This dataset was expanded to 9,000 images using the 15 different preprocessing and augmentation techniques to enhance feature diversity and model generalisation. Various Convolutional Neural Network (CNN) architectures, including the conventional and pre-trained models, were trained and evaluated. The highest performance was achieved by our proposed model, a hybrid architecture integrating DenseNet121-based CNN for the feature extraction with an XGBoost classifier, achieving 92.00% accuracy with strong precision, recall, and F1 scores. To ensure practical utility, the final model was deployed as a user-friendly web application with the name FasalSaathi, providing the farmers and agricultural professionals with a low-cost, accessible tool for real-time leaf health diagnosis. The proposed dataset serves as a new benchmark for the future.

Keywords: Indian Ridge Gourd, Plant Disease Detection, Hybrid Deep Learning, Convolutional Neural Networks, XGBoost, Dataset Generation, Smart Agriculture.

1 Introduction

Agriculture is crucial to the economies of many countries, especially those that are still developing. It also plays an important role in ensuring global food security. A noteworthy challenge that poses a direct threat to crop yield and quality is that diseases can appear quickly and sometimes go unnoticed at first. In the past, identifying these diseases has been dependent on manual inspection by farmers or agricultural experts. [7] [26] This way of working is time-consuming, expensive, and reliant on personal judgment, often causing delayed detection, misdiagnosis, and major economic losses for farmers. The lack of a scalable and timely diagnostic system for large-scale farming operations highlights the need for an improved solution. The recent advances in Artificial Intelligence (AI) and particularly Deep Learning (DL) [2] [19] [22] have provided another way to address these challenges. Convolutional Neural Networks (CNNs) have proven to be highly effective in analysing visual data and making them well-suited for image-based plant disease detection. Over the past decade, many studies have shown their ability to identify plant diseases using images of leaves, stems, and fruits. [17] [20].

While the use of deep learning has grown, most studies so far have focused on well-documented crops such as tomato, potato, and grape. [15] [19] [23] In contrast, many of the locally grown crops do not have dedicated research or a major dataset made for them. The current study was carried out to address this specific research gap by focusing on Indian Ridge gourd, a crop that is locally grown [1] in the northern and southern parts of India. Ridge Gourd is a commercial crop grown in various parts of India like Karnataka, Tamil Nadu, Andhra Pradesh, Maharashtra and Rajasthan [1]. We interacted with the farmers in rural Jaipur, where they told us that the diseases often go unnoticed until they have spread severely. Studies show that diseases can cause 20-50% of yield loss, and even up to 70% in severe cases, making the early detection of the diseases very crucial [2]. The work covers the full AI process, from the creation of a real-world image dataset to developing an effective hybrid deep learning model. The current study aims to present an automated system for detecting Indian Ridge gourd leaf disease and to create a resource that can support future research and applications.

In recent years, there has been a rapid growth in the research on deep learning for plant disease detection, with many studies and collaborations taking place. The methods and results from this work provide a crucial background for the present study. An analysis of research from 2014 to 2024 shows a clear shift toward using deep learning, especially CNNs, as the main approach for image-based plant disease detection. Unlike traditional machine learning methods, which rely on manual feature extraction, these models can automatically learn detailed features from raw images. Studies show that many CNN models, such as AlexNet, VGGNet, ResNet, InceptionNet, and DenseNet [10] [24] [29], use a technique called Transfer Learning to adapt pre-trained models for agricultural tasks. Along with that, techniques like data augmentation are also widely used to expand small datasets, improving the reliability and generalisation of models. Together,

these advanced architectures and data strategies represent a collective effort to achieve higher accuracy and performance in automated disease detection.

Most research on plant disease detection has focused on a few well-studied crops like tomato, potato, and grape, mainly because a curated large dataset exists for them. This has created a cycle- where the datasets attract more studies, which in turn leads to more accurate models. In contrast, many local crops lack a dedicated dataset or research done for them, creating a challenge to use an AI-based solution for the farmers who grow them. After a review of existing studies, open data sources confirmed that no dedicated dataset exists for the Indian Ridge gourd diseases, and the crop is absent from the major keyword network. To fulfil this gap, the present study introduces a complete pipeline for Ridge gourd disease detection, starting from field data collection to developing a hybrid deep learning model and deploying a practical application.

The major contributions of the study are summarised as follows:

- Making a real-world dataset on Ridge gourd leaf diseases in India to fill a gap in the research.
- Development of a hybrid deep learning architecture (DenseNet121-XGBoost) that achieved a notable classification accuracy of 92.00%.
- Successful implementation of a comprehensive AI pipeline, delivering an easy-to-use, low-cost solution to help Indian farmers improve crop yield and sustainability.

The outcome of the current study contributes to research and provides a practical solution for improving agricultural sustainability and productivity. The work was carried out in three major steps: building a dataset for a crop that has not been studied before, developing a new hybrid deep learning model, and creating a real-world practical application to assist farmers.

2 Methodology

The methodology for this work was designed to be a complete, end-to-end pipeline, to maintain accuracy and real-world usefulness. The process began with on-ground data collection, followed by extensive preprocessing, model development, and final deployment. The overall research workflow is summarised in Figure 1, providing a visual roadmap for the whole process.

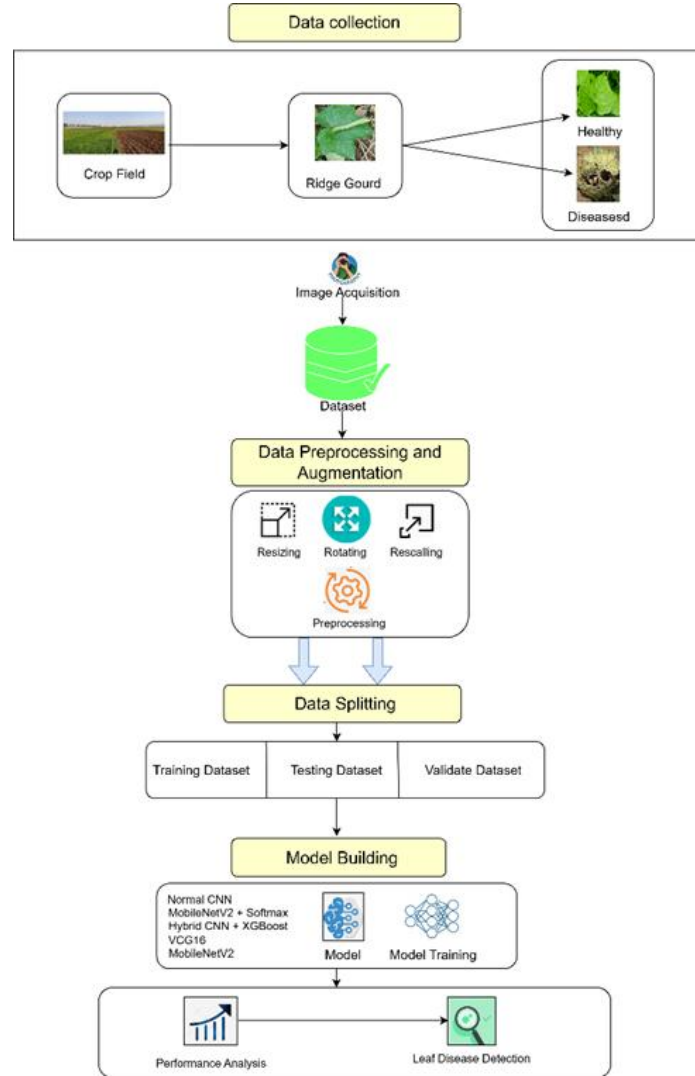


Fig. 1. Research flow diagram adopted in the present study

2.1 Dataset Selection

To overcome the absence of a publicly available dataset for Indian Ridge gourd leaf diseases, a new dataset was created from scratch. [1] A total of 600 raw images of Indian Ridge gourd leaves were collected from local farms in Vidhani, Jaipur, using a smartphone camera. The images were captured under different natural lighting and farm conditions to ensure the final dataset represents real field conditions and would not be biased by ideal, lab-quality settings. The images were then sorted into three categories manually: [1]

- Healthy leaves: 300 images
- Leaf Minor Infestation: 125 images
- Mosaic Virus: 175 images

The number of images for each disease type was determined based on how often they appeared during data collection, ensuring the dataset reflects real farm conditions. Figure 2 shows sample images from the dataset [1] to give an idea of each category.



Fig. 2. Sample images from the dataset (showing healthy leaves, leaves with Leaf Miner Infestation, and leaves affected by Mosaic Virus)

2.2 Research Methodology

Several comparisons were carried out by training and evaluating a variety of CNN-based models on the proposed dataset. The models were selected to cover both simple and advanced models, from a basic conventional CNN to high-performance pre-trained architectures. This included a Normal CNN, MobileNetV2 with a SoftMax classifier, VGG16, MobileNetV2, and our proposed Hybrid CNN-XGBoost model. Testing this range of models allowed us to check whether our approach performed better than the standard methods.

2.3 Algorithm Development and Deep Learning Theory

The main focus of the current study involved the evaluation of multiple deep learning models, leading to the design of a high-performance hybrid model. A CNN operates by passing an input image through a series of convolution, activation, and pooling layers to automatically learn hierarchical features, from simple edges and textures in the initial layers to complex patterns and objects in deeper layers. This process eliminates the need for manual feature engineering. XGBoost (Extreme Gradient Boosting) [16] [19], on the other hand, is a powerful and optimized version of the gradient-boosted decision tree algorithm. It is effective at handling complex, non-linear relationships by building a series of decision trees, with each new tree improving on the mistakes of the earlier ones. Our proposed model followed a hybrid approach that combines deep learning with gradient boosting. The first phase of our proposed model used a DenseNet121 CNN as a fixed-feature extractor. The DenseNet121 backbone, pre-trained on ImageNet, is known for its efficiency in capturing useful image features. The final classification layers of the CNN were dropped, and the output from its bottleneck layers was used as a feature vector. This vector was then passed to the second component of

the hybrid model: an XGBoost classifier. By providing XGBoost with the automatically extracted feature vectors from the CNN, the hybrid model took advantage of the representational strength of deep learning to transform the image data into a well-structured, tabular-like format. XGBoost then utilised its strong discriminative ability to generate the final, highly accurate prediction. This approach separates feature extraction from classification, a strategy shown to be highly effective in complex classification tasks. The overall design of the model is shown in Figure 3.

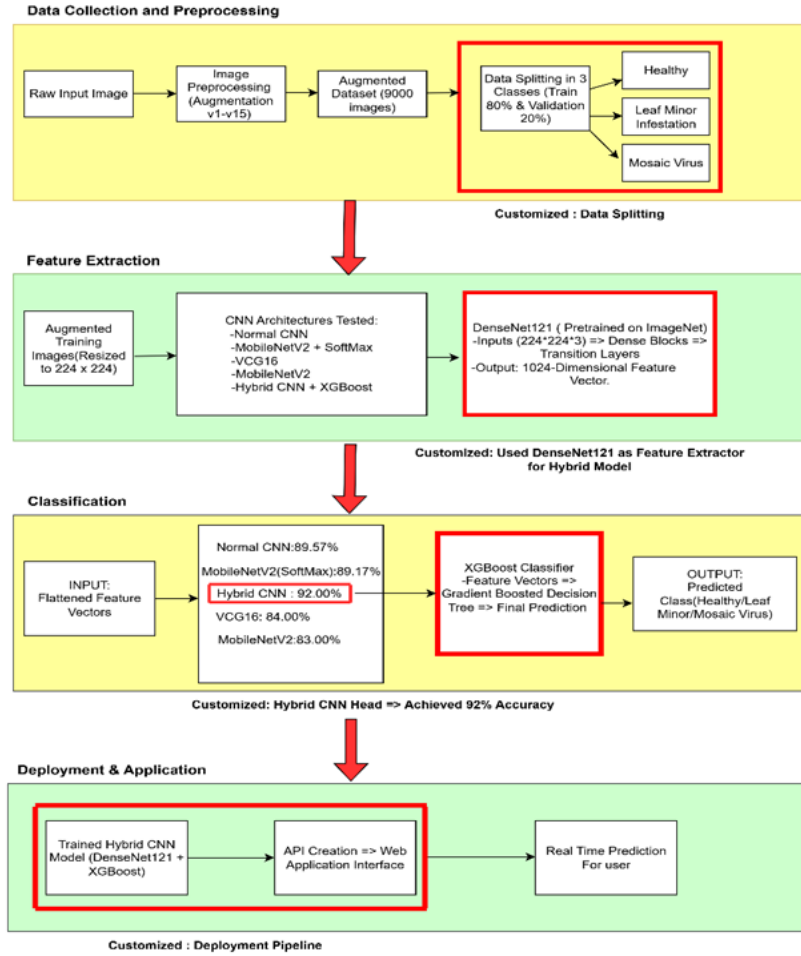


Fig. 3. Proposed model architecture.

2.4 Preprocessing

The initial dataset of 600 ridge gourd leaf images (300 Healthy, 175 Mosaic Virus, 125 Leaf Minor Infestation) was insufficient for training a deep learning model that generalises well to new images. To overcome this limitation, a series of 15 different image preprocessing and augmentation techniques was applied to improve generalisation and

overfitting [5] [9] [28]. These techniques expanded the dataset and introduced more variety, helping the model handle variations in lighting, orientation, and subtle feature details. The techniques included resizing, grayscale conversion, binary thresholding, CLAHE, Gaussian blur, median blur, bilateral filtering, adaptive mean and adaptive Gaussian thresholding, rotation, horizontal and vertical flips, brightness/contrast shifts, and mild noise injection. Each original image was transformed through 15 variants, producing a final dataset of 9,000 images [1], ensuring higher variability and robustness for training.

In the current study, all the preprocessing and augmentation were performed offline to expand the dataset size from 600 to 9000 images. After generating the augmented dataset, a split of 70-15-15 of train-test-validate was applied. A similar augmentation-before-splitting workflow has been reported in recent literature, including the work of Bhagat et al. [5], who employed offline augmentation followed by a random dataset split for plant disease classification. We acknowledge that splitting after augmentation is not ideally used due to the limitation of getting the data leakage possibility. However, all the models were trained and evaluated using the same consistent methodology, ensuring fair comparison. Future work will implement group-wise splitting to avoid potential leakage effects.

3 Results and Discussion

Careful evaluation of all models confirmed the effectiveness of our proposed hybrid architecture and gave useful insights into how different deep learning techniques perform on a real-world agricultural dataset.

3.1. Comparative Performance Analysis

The models were compared based on a standard set of evaluation metrics, including accuracy, precision, recall, and F1-score. Table 1 summarises their performance, clearly showing that our proposed Hybrid CNN + XGBoost model outperforms the others.

Table 1. Performance Comparison of Evaluated Deep Learning Models on the Plant Disease Dataset

Model	Accuracy (%)	Precision	Recall	F1 Score
Normal CNN	89.57	0.88	0.89	0.88
MobileNetV2 + Soft-Max	89.17	0.87	0.87	0.86
VGG16	84.00	0.81	0.79	0.80
MobileNetV2	83.00	0.83	0.83	0.83
Hybrid CNN + XGBoost	92.00	0.92	0.92	0.92

Our proposed model achieved an accuracy of 92.00% which is slightly better than the next best performing model [11] [14] [25] of Normal CNN with 89.57% accuracy. The

performance gap between the best-performing model and the other models shows a clear advantage of combining a feature extractor with a powerful classifier. While models like VGG16 (84.00%) and MobileNetV2 (83.00%) are good at extracting the features, their performance with a basic classifier suggests that a more robust classification approach is needed for this specific task. A conventional CNN's final layer, usually a SoftMax classifier, performs linear classification on the extracted features. In contrast, XGBoost, a powerful decision-tree-based model, can capture more complex decision rules [11] [17] from the same feature vector provided by the CNN. This ability to model complex feature relationships enables our proposed hybrid model to achieve higher accuracy and better results across all metrics (Precision, Recall, F1 Score) compared to end-to-end CNNs. These performance differences are shown in Figures 4 [18] [21]. This comparative evaluation includes multiple CNN baselines (Normal CNN, Custom CNN, VGG16, MobileNetV2) and also a hybrid model to ensure that a fair and comprehensive assessment of the proposed approach is met.



Fig. 4. Comparison of model performance metrics (Accuracy, Precision, Recall, F1-score)

3.2 Performance Evaluation and Practical Implications

Our model performance was further analysed using ROC curves and a confusion matrix. The ROC curve showed a clear separation between healthy and diseased leaves, with the highest AUC compared to other models (Figure 5). The confusion matrix revealed that most errors occurred between similar diseases, but healthy leaves were almost always correctly identified (Figure 6).

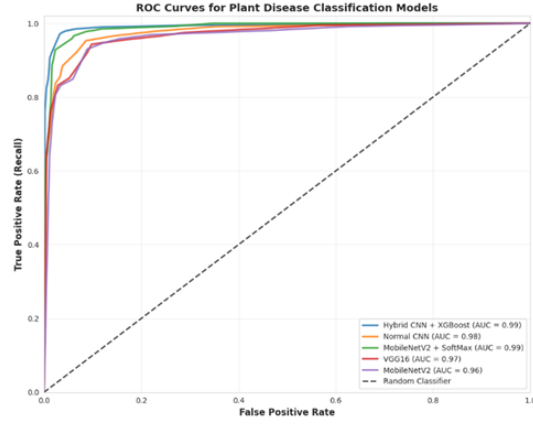


Fig. 5. ROC Curve Comparison

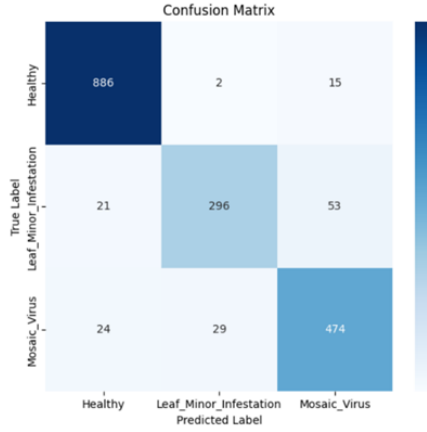


Fig. 6. Confusion Matrix of Proposed Model

The statistical test (Friedman with Nemenyi post hoc) confirmed that the proposed Hybrid CNN + XGBoost model is significantly better than the others (Figure 7).

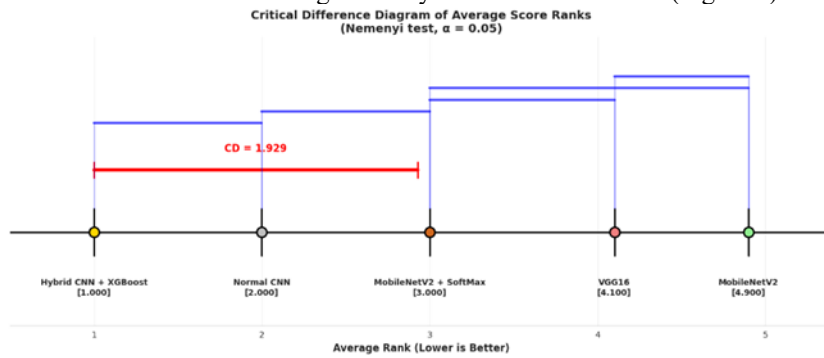


Fig. 7. Nemenyi post hoc Test

Beyond accuracy (92%), this model offers real-world value by enabling early disease detection and reducing crop losses. We deployed it as an easy-to-use web application, trained on real-world data for reliable field use.

4 Conclusion

4.1 Conclusion

The current study presents a complete AI solution for detecting Indian Ridge gourd leaf diseases, an important and unexplored area. We first created a dedicated dataset for this crop and developed a new Hybrid CNN + XGBoost model, which achieved a high accuracy of 92%. The comparison showed that our proposed model outperformed other models significantly in all the evaluation parameters. A web application named FasalSaathi was also built to make the solution easily accessible to the farmers and other research experts. This work lays a foundation for future smart farming applications.

The hybrid model and the deployed web application were qualitatively validated using real field images to ensure practical usability, and a more extensive statistical significance analysis is planned for future work.

4.2 Future Scope and Recommendation

This work opens several opportunities for future improvements:

- **Extend the Proposed Dataset:** Expand the dataset to include more disease types, growth stages, and varied environments to help the model work better.
- **Implement with IoT Devices:** Embed the model in mobile cameras or IoT sensors for on-field use.
- **Develop a Cross-Crop System:** Apply the same approach to other local crops like brinjal and bottle gourd.

These steps aim to make the system more scalable and useful in real farming situations.

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